

第四章作业参考答案

4.12 考虑下面傅里叶变换对:

$$e^{-|t|} \xleftrightarrow{\mathcal{F}} \frac{2}{1+\omega^2}.$$

(a) 利用恰当的傅里叶变换性质求 $te^{-|t|}$ 的傅里叶变换.

由频域微分性质, 有

$$te^{-|t|} \xleftrightarrow{\mathcal{F}} j \frac{d}{d\omega} \frac{2}{1+\omega^2} = -\frac{4j\omega}{(1+\omega^2)^2}.$$

(b) 根据 (a) 的结果, 再结合对偶性质, 求

$$\frac{4t}{(1+t^2)^2}$$

的傅里叶变换.

$$\begin{aligned} -\frac{4jt}{(1+t^2)^2} &\xleftrightarrow{\mathcal{F}} 2\pi\omega e^{-|\omega|}, \\ \Rightarrow \frac{4t}{(1+t^2)^2} &\xleftrightarrow{\mathcal{F}} 2\pi j\omega e^{-|\omega|}. \end{aligned}$$

4.21 求下列每一信号的傅里叶变换:

(a) $[e^{-\alpha t} \cos \omega_0 t] u(t), \alpha > 0$

$$\begin{aligned} &[e^{-\alpha t} \cos \omega_0 t] u(t) \\ &= \frac{1}{2} e^{-(\alpha-j\omega_0)t} u(t) + \frac{1}{2} e^{-(\alpha+j\omega_0)t} u(t) \\ &\xleftrightarrow{\mathcal{F}} \frac{1}{2(\alpha-j\omega_0+j\omega)} - \frac{1}{2(\alpha+j\omega_0+j\omega)}. \end{aligned}$$

(b) $e^{-3|t|} \sin 2t$.

$$\begin{aligned} &e^{-3|t|} \sin 2t \\ &= \frac{1}{j2} e^{-3t} e^{j2t} u(t) - \frac{1}{j2} e^{-3t} e^{-j2t} u(t) + \frac{1}{j2} e^{3t} e^{j2t} u(-t) - \frac{1}{j2} e^{3t} e^{-j2t} u(-t) \\ &\xleftrightarrow{\mathcal{F}} \frac{1}{j2(3-j2+j\omega)} - \frac{1}{j2(3+j2+j\omega)} - \frac{1}{j2(-3-j2+j\omega)} + \frac{1}{j2(-3+j2+j\omega)} \\ &= \frac{1}{j6+4-2\omega} - \frac{1}{j6-4-2\omega} + \frac{1}{j6-4+2\omega} - \frac{1}{j6+4+2\omega} \\ &= -\frac{j3}{9+(\omega-2)^2} + \frac{j3}{9+(\omega+2)^2}. \end{aligned}$$

$$(c) \quad x(t) = \begin{cases} 1 + \cos \pi t & , |t| \leq 1 \\ 0 & , |t| > 1 \end{cases}.$$

易知 $x(t)$ 为实值偶函数, 因此 $X(j\omega)$ 为实值偶函数.

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \\ &= \int_{-1}^1 x(t)e^{-j\omega t} dt \\ &= \int_{-1}^1 x(t) \cos \omega t dt \quad (\text{由实值性质可以去除必定积分结果为纯虚数的项.}) \\ &= 2 \int_0^1 \cos \omega t + \cos \pi t \cos \omega t dt \\ &= \frac{2 \sin \omega}{\omega} + \frac{\sin(\omega + \pi)}{\pi + \omega} + \frac{\sin(\omega - \pi)}{\omega - \pi} \\ &= \frac{2 \sin \omega}{\omega} - \frac{\sin(\omega)}{\omega + \pi} - \frac{\sin(\omega)}{\omega - \pi} \\ &= \frac{2 \sin \omega}{\omega} - \frac{2 \omega \sin(\omega)}{\omega^2 - \pi^2} \end{aligned}$$

$$(d) \quad \sum_{k=0}^{\infty} \alpha^k \delta(t - kT) \quad , |\alpha| < 1.$$

$$\begin{aligned} \mathcal{F} \left[\sum_{k=0}^{\infty} \alpha^k \delta(t - kT) \right] (j\omega) &= \int_{-\infty}^{+\infty} \left(\sum_{k=0}^{\infty} (\alpha^k \delta(t - kT)) e^{-j\omega t} \right) dt \\ &= \sum_{k=0}^{+\infty} \int_{-\infty}^{+\infty} \alpha^k \delta(t - kT) e^{-j\omega t} dt \\ &= \sum_{k=0}^{+\infty} (\alpha e^{-j\omega T})^k \\ &= \frac{1}{1 - \alpha e^{-j\omega T}}. \end{aligned}$$

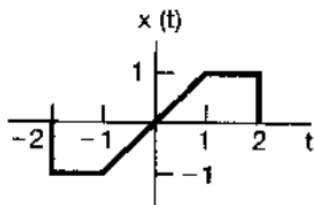
$$(e) \quad [te^{-2t} \sin 4t] u(t).$$

$$\begin{aligned} &[te^{-2t} \sin 4t] u(t) \\ &= \frac{1}{j2} te^{-2t} e^{j4t} u(t) - \frac{1}{j2} te^{-2t} e^{-j4t} u(t) \\ &\xleftrightarrow{\mathcal{F}} \frac{1}{2} \frac{d \left(\frac{1}{2-j4+j\omega} - \frac{1}{2+j4+j\omega} \right)}{d\omega} \\ &= \frac{1}{j2} \left(\frac{1}{(2-j4+j\omega)^2} - \frac{1}{(2+j4+j\omega)^2} \right). \end{aligned}$$

$$(f) \left[\frac{\sin \pi t}{\pi t} \right] \left[\frac{\sin 2\pi(t-1)}{\pi(t-1)} \right].$$

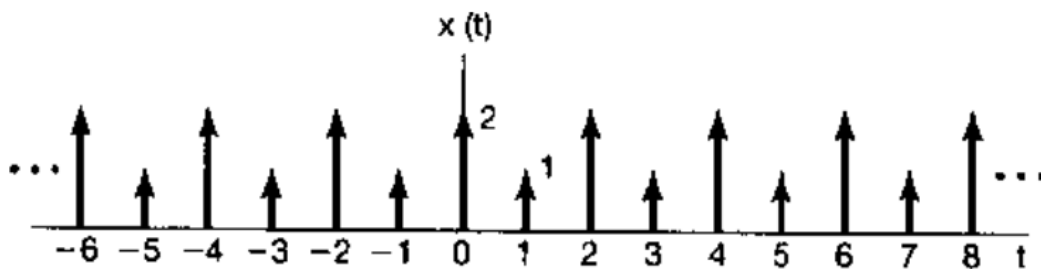
$$\begin{aligned} & \left[\frac{\sin \pi t}{\pi t} \right] \left[\frac{\sin 2\pi(t-1)}{\pi(t-1)} \right] \\ &= 2 \text{Sa}(\pi t) \text{Sa}(2\pi(t-1)) \\ &\stackrel{\mathcal{F}}{\longleftrightarrow} \frac{1}{\pi} [u(-\omega + \pi) - u(-\omega - \pi)] * [e^{-j\omega} (u(-\omega + 2\pi) - u(-\omega - 2\pi))] \\ &= \frac{1}{\pi} [u(\omega + \pi) - u(\omega - \pi)] * [e^{-j\omega} (u(\omega + 2\pi) - u(\omega - 2\pi))] \\ &= \begin{cases} \frac{1}{j\pi} (e^{-j\omega} + 1) & , -3\pi \leq \omega < -\pi \\ -\frac{1}{j\pi} (e^{-j\omega} + 1) & , \pi < \omega \leq 3\pi \\ 0 & , else \end{cases} \end{aligned}$$

(g) $x(t)$ 如图所示.



$$\begin{aligned} x(t) &= \begin{cases} -1 & , -2 \leq t < -1 \\ t & , -1 \leq t \leq 1 \\ 1 & , 1 < t \leq 2 \\ 0 & , else \end{cases} \\ &\stackrel{\mathcal{F}}{\longleftrightarrow} \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-2}^2 x(t) e^{-j\omega t} dt \\ &= - \int_{-2}^{-1} e^{-j\omega t} dt + \int_{-1}^1 t e^{-j\omega t} dt + \int_1^2 e^{-j\omega t} dt \\ &= \frac{1}{\omega} \left(j2 \cos 2\omega - \frac{j2}{\omega} \sin \omega \right) \\ &= \frac{j2}{\omega} \left(\cos 2\omega - \frac{\sin \omega}{\omega} \right). \end{aligned}$$

(h) $x(t)$ 如图所示.



令 $z(t, T) = \sum_{n=-\infty}^{+\infty} \delta(t - T * n)$, 则

$$x(t) = z(t, 1) - z(t, 2) + 2z(t, 2) = z(t, 1) + z(t, 2).$$

$$\therefore z(t, T) \xleftrightarrow{\mathcal{F}} \frac{2\pi}{T} \sum_{-\infty}^{\infty} \delta\left(\omega - \frac{2n\pi}{T}\right),$$

$$\therefore x(t) \xleftrightarrow{\mathcal{F}} 2\pi \sum_{-\infty}^{\infty} \delta(\omega - 2n\pi) + \pi \sum_{-\infty}^{\infty} \delta(\omega - n\pi).$$

(i) $x(t) = \begin{cases} 1 - t^2, & 0 < t < 1 \\ 0, & \text{其余 } t \end{cases}.$

$$\begin{aligned} x(t) &= \begin{cases} 1 - t^2, & 0 < t < 1 \\ 0, & \text{其余 } t \end{cases} \\ &\xleftrightarrow{\mathcal{F}} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \int_0^1 (1 - t^2) e^{-j\omega t} dt \\ &= \frac{1}{j\omega} + \frac{2}{j\omega^3} - \frac{2e^{-j\omega}}{\omega^2} - \frac{2e^{-j\omega}}{j\omega^3} \end{aligned}$$

(j) $\sum_{n=-\infty}^{+\infty} e^{-|t-2n|}.$

令 $z(t) = e^{-|t|}$, 则 $x(t) = z(t) * \delta_T(t).$

$$\begin{aligned} Z(j\omega) &= \frac{2}{\omega^2 + 1}, \\ \Rightarrow X(j\omega) &= \frac{2}{\omega^2 + 1} \pi \sum_{n=-\infty}^{+\infty} \delta(\omega - n\pi) = \frac{2\pi}{\omega^2 + 1} \sum_{n=-\infty}^{+\infty} \delta(\omega - n\pi). \end{aligned}$$

4.22 对下列每一个变换求对应的连续时间信号:

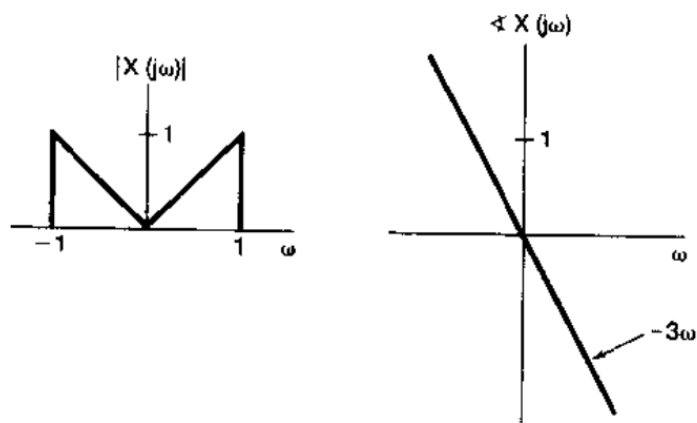
(a) $X(j\omega) = \frac{2 \sin[3(\omega - 2\pi)]}{\omega - 2\pi}.$

$$\begin{aligned} X(j\omega) &= 6 \cdot \frac{\sin[3(\omega - 2\pi)]}{3(\omega - 2\pi)} \\ &= 6 \text{Sa}(3(\omega - 2\pi)) \\ &\xleftrightarrow{\mathcal{F}^{-1}} (u(t+3) - u(t-3)) e^{j2\pi t}. \end{aligned}$$

(b) $X(j\omega) = \cos(4\omega + \pi/3)$.

$$\begin{aligned} X(j\omega) &= \frac{1}{2} (e^{j(4\omega + \pi/3)} + e^{-j(4\omega + \pi/3)}) \\ &= \frac{1}{2} e^{j4\omega} e^{j\pi/3} + \frac{1}{2} e^{-j4\omega} e^{-j\pi/3} \\ &\xleftrightarrow{\mathcal{F}^{-1}} \frac{1}{2} e^{j\pi/3} \delta(t+4) + \frac{1}{2} e^{-j\pi/3} \delta(t-4). \end{aligned}$$

(c) $X(j\omega)$ 的模和相位如图所示.

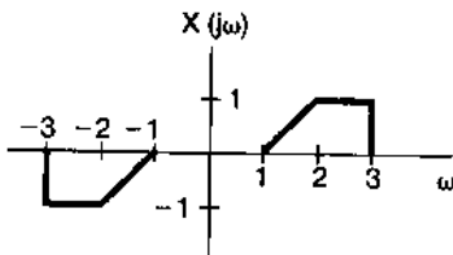


$$\begin{aligned} X(j\omega) &= \begin{cases} -\omega e^{-j3\omega} & , -1 \leq \omega \leq 0 \\ \omega e^{-j3\omega} & , 0 \leq \omega \leq 1 \\ 0 & , else \end{cases} \\ &\xleftrightarrow{\mathcal{F}^{-1}} \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \left(- \int_{-1}^0 \omega e^{-j3\omega} e^{j\omega t} d\omega + \int_0^1 \omega e^{-j3\omega} e^{j\omega t} d\omega \right) \\ &= \frac{1}{\pi} \left(\frac{\sin(t-3)}{t-3} + \frac{\cos(t-3)-1}{(t-3)^2} \right). \end{aligned}$$

(d) $X(j\omega) = 2[\delta(\omega-1) - \delta(\omega+1)] + 3[\delta(\omega-2\pi) + \delta(\omega+2\pi)]$.

$$x(t) = \frac{j2}{\pi} \sin t + \frac{3}{\pi} \cos 2\pi t.$$

(e) $X(j\omega)$ 如图所示.



$$\begin{aligned}
2\pi x(t) &= -\int_{-3}^{-2} e^{j\omega t} d\omega + \int_{-2}^{-1} (\omega+1)e^{j\omega t} d\omega + \int_1^2 (\omega-1)e^{j\omega t} d\omega + \int_2^3 e^{j\omega t} d\omega \\
&= \frac{1}{jt} (e^{-j3t} - e^{-j2t} + e^{-jt} - e^{-j2t} + e^{jt} - e^{j2t} + e^{j3t} - e^{j2t}) \\
&\quad + \frac{1}{j^2 t^2} (2jte^{-2jt} - jte^{-jt} - e^{-jt} + e^{-2jt} - jte^{jt} + 2jte^{2jt} - e^{2jt} + e^{jt}) \\
&= \frac{2\cos 3t + 2\frac{\sin t}{t} - 2\frac{\sin 2t}{t}}{jt}. \\
\therefore x(t) &= \frac{\cos 3t}{j\pi t} + \frac{\sin t - \sin 2t}{j\pi t^2}.
\end{aligned}$$

4.25 设 $X(j\omega)$ 为图 1 信号 $x(t)$ 的傅里叶变换:

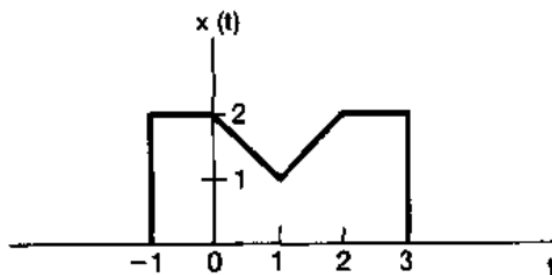


Figure 1: $x(t)$ 波形.

(a) 求 $\angle X(j\omega)$.

令 $y(t) = x(t+1)$, 可知 $y(t)$ 是实偶函数, 因而其傅立叶变换的相位 $\angle Y(j\omega) = 0$.

$$\therefore X(j\omega) = Y(j\omega)e^{-j\omega},$$

$$\therefore \angle X(j\omega) = \angle Y(j\omega) - \omega = -\omega.$$

(b) 求 $X(j0)$.

$$\begin{aligned}
X(j0) &= \int_{-\infty}^{+\infty} x(t)e^{-j0t} dt \\
&= \int_{-1}^3 x(t) dt \\
&= 7.
\end{aligned}$$

(c) 求 $\int_{-\infty}^{\infty} X(j\omega) d\omega$.

$$\begin{aligned}
&\int_{-\infty}^{\infty} X(j\omega) d\omega \\
&= 2\pi \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \right] \Big|_{t=0} \\
&= 2\pi x(0) \\
&= 4\pi.
\end{aligned}$$

(d) 计算 $\int_{-\infty}^{\infty} X(j\omega) \frac{2\sin \omega}{\omega} e^{j2\omega} d\omega$.

$$\begin{aligned}
& \int_{-\infty}^{\infty} X(j\omega) \frac{2 \sin \omega}{\omega} e^{j2\omega} d\omega \\
&= \int_{-\infty}^{\infty} X(j\omega) \text{Sa}(\omega) e^{j2\omega} d\omega \\
&= 2\pi [x(t) * (u(t+1) - u(t-1))] |_{t=2} \\
&= 2\pi \cdot 3.5 \\
&= 7\pi.
\end{aligned}$$

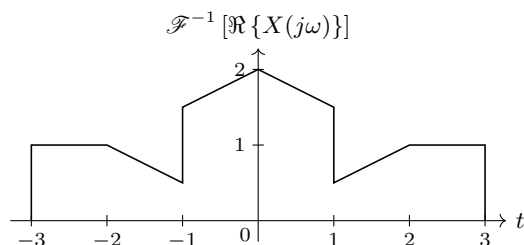
(e) 计算 $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$.

$$\begin{aligned}
& \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \\
&= \int_{-\infty}^{\infty} |Y(j\omega) e^{j\omega}|^2 d\omega \\
&= \int_{-\infty}^{\infty} Y(j\omega)^2 d\omega \\
&= 2\pi [y(t) * y(t)] |_{t=0} \\
&= 2\pi \int_{-2}^2 y^2(t) dt \\
&= \frac{76}{3} \pi.
\end{aligned}$$

(f) 画出 $\Re\{X(j\omega)\}$ 的反变换.

$$\text{有 } \Re\{X(j\omega)\} \xleftrightarrow{\mathcal{F}^{-1}} x_e(t) = \frac{x(t) + x(-t)}{2}.$$

其图像如下图.



注意: 不必具体算出 $X(j\omega)$ 而能完成以上全部计算.

4.28 (a) 设 $x(t)$ 有傅里叶变换 $X(j\omega)$, 令 $p(t)$ 为基波频率 ω_0 的周期信号, 其傅里叶级数表示是

$$p(t) = \sum_{n=-\infty}^{+\infty} a_n e^{jn\omega_0 t}.$$

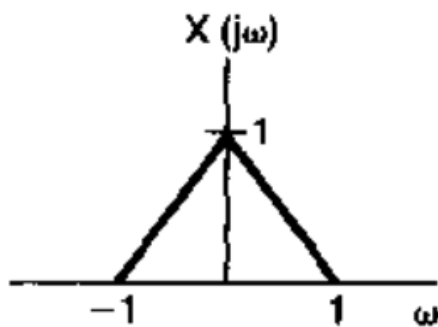
求

$$y(t) = x(t)p(t) \quad (\star)$$

的傅里叶变换表示式.

$$\begin{aligned}
Y(j\omega) &= \frac{1}{2\pi} X(j\omega) * P(j\omega) \\
&= \frac{1}{2\pi} X(j\omega) * \left(\sum_{n=-\infty}^{+\infty} 2\pi a_n \delta(\omega - n\omega_0) \right) \\
&= \sum_{n=-\infty}^{+\infty} a_n X(j(\omega - n\omega_0)).
\end{aligned}$$

(b) 设 $X(j\omega)$ 如图所示没对下列每一个 $p(t)$ 画出式 (*) 中 $y(t)$ 的频谱:

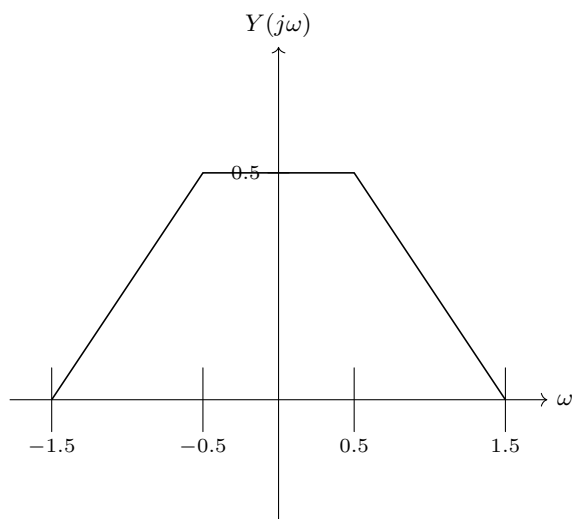


$$X(j\omega) = \begin{cases} \omega + 1 & , -1 \leq \omega \leq 0 \\ 1 - \omega & , 0 \leq \omega \leq 1 \\ 0 & , else \end{cases}$$

(1) $p(t) = \cos(t/2)$.

$$\begin{aligned} Y(j\omega) &= \frac{1}{2\pi} X(j\omega) * P(j\omega) \\ &= X(j\omega) * [\delta(2\omega - 1) + \delta(2\omega + 1)] \\ &= \frac{1}{2} X(j\omega) * \left[\delta\left(\omega + \frac{1}{2}\right) - \delta\left(\omega - \frac{1}{2}\right) \right] \\ &= \frac{1}{2} \left[X\left(j\left(\omega - \frac{1}{2}\right)\right) + X\left(j\left(\omega + \frac{1}{2}\right)\right) \right] \\ &= \begin{cases} \frac{1}{2}x + \frac{3}{4} & , -\frac{3}{2} \leq x < -\frac{1}{2} \\ \frac{1}{2} & , -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \frac{3}{4} - \frac{1}{2}x & , \frac{1}{2} < x \leq \frac{3}{2} \\ 0 & , else \end{cases} \end{aligned}$$

频谱图如下.

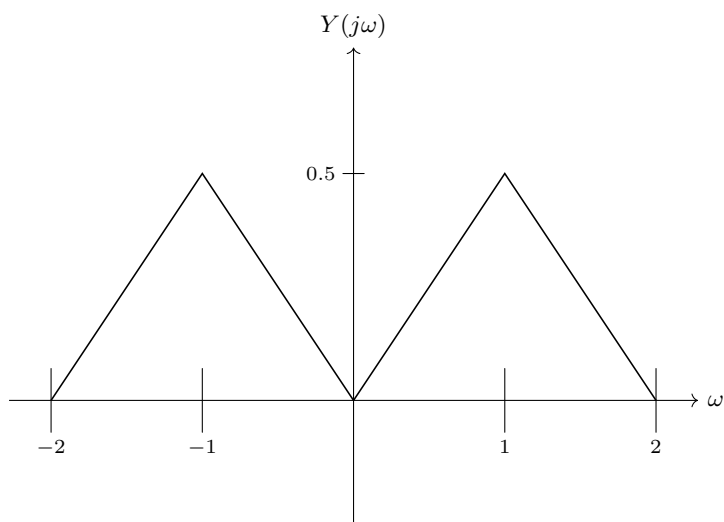


(2) $p(t) = \cos t$.

由 (1) 易知,

$$Y(j\omega) = \frac{1}{2} [X(j(\omega - 1)) + X(j(\omega + 1))].$$

频谱图如下.

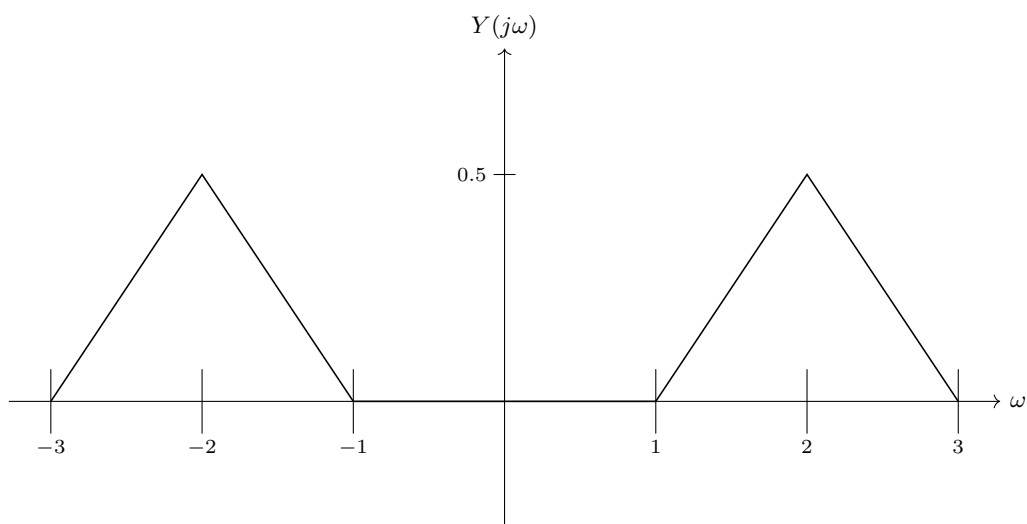


(3) $p(t) = \cos 2t$.

同理,

$$Y(j\omega) = \frac{1}{2} [X(j(\omega - 2)) + X(j(\omega + 2))].$$

频谱图如下.

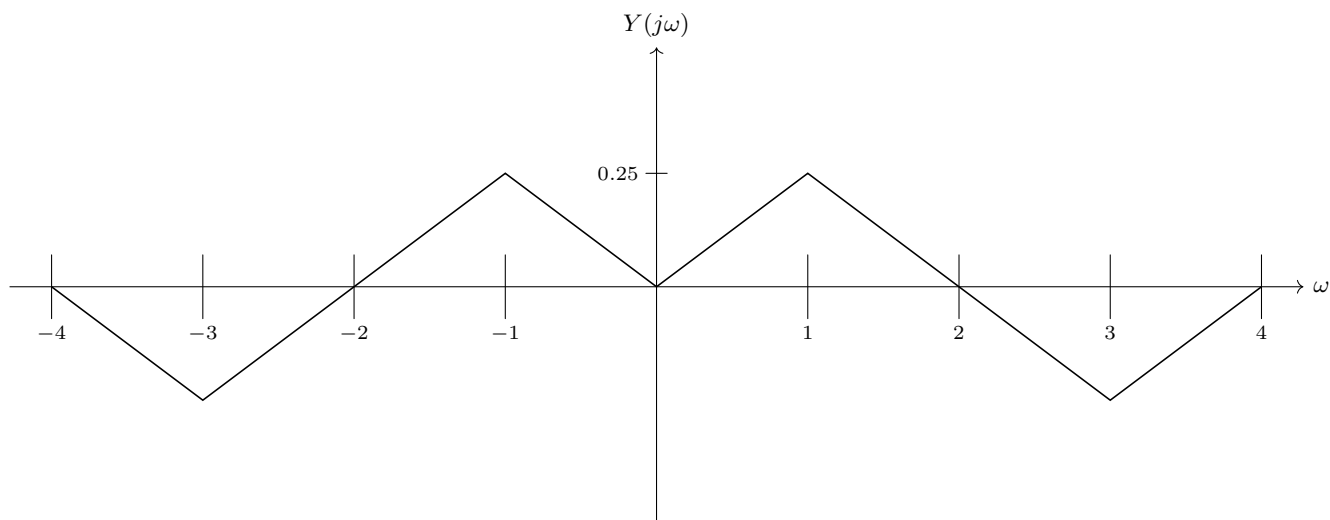


(4) $p(t) = (\sin t)(\sin 2t)$.

$$p(t) = \frac{1}{2} \cos t - \frac{1}{2} \cos 3t.$$

$$Y(j\omega) = \frac{1}{4} [X(j(\omega - 1)) + X(j(\omega + 1)) - X(j(\omega - 3)) - X(j(\omega + 3))].$$

频谱图如下.

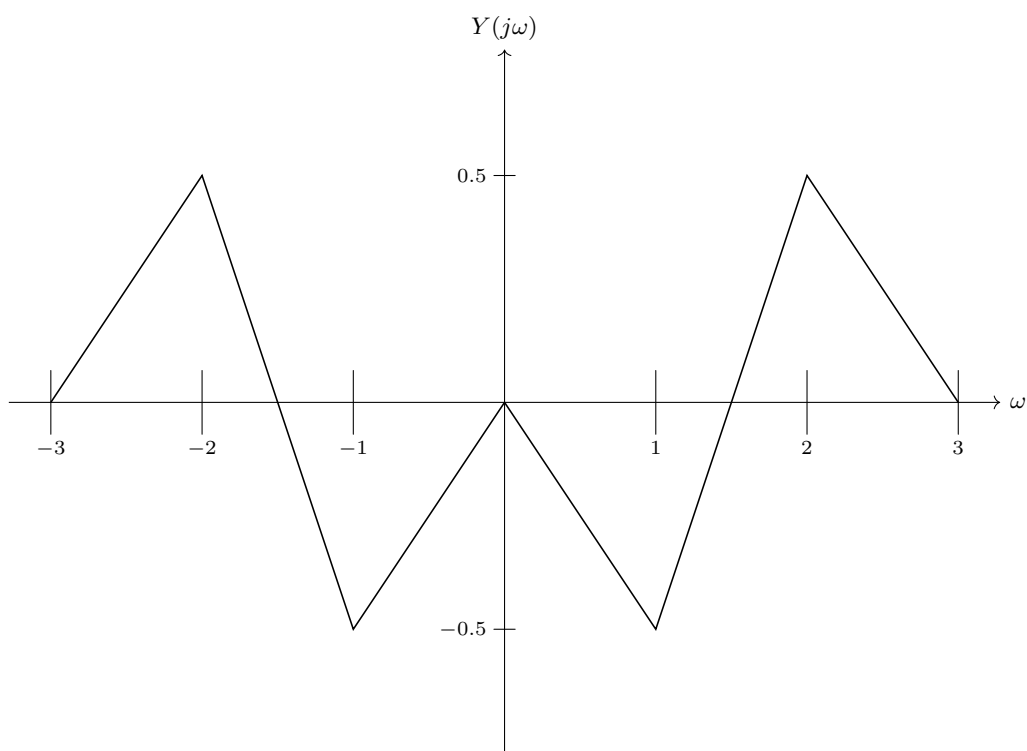


(5) $p(t) = \cos 2t - \cos t$.

同理,

$$Y(j\omega) = \frac{1}{2} [X(j(\omega - 2)) + X(j(\omega + 2)) - X(j(\omega - 1)) - X(j(\omega + 1))].$$

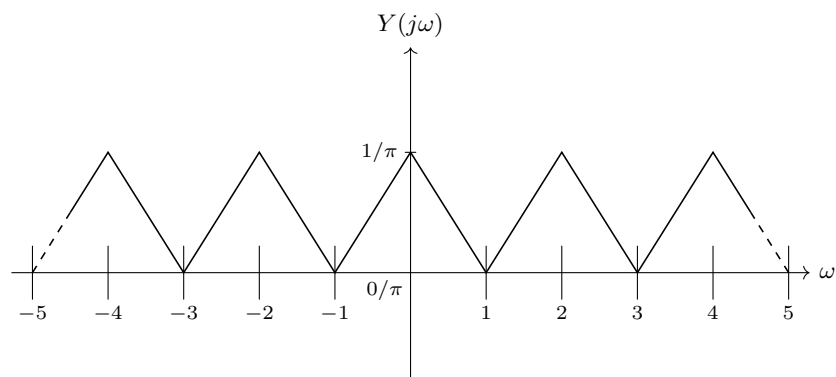
频谱图如下.



(6) $p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - \pi n)$.

$$P(j\omega) = 2 \sum_{n=-\infty}^{+\infty} \delta(\omega - 2n).$$

频谱图如下.

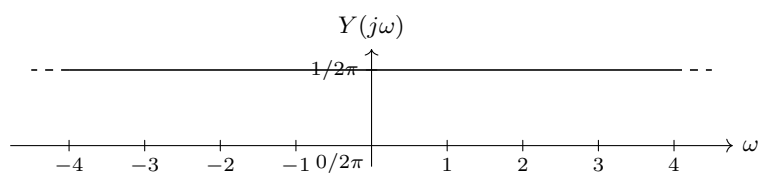


$$(7) \quad p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - 2\pi n).$$

$$P(j\omega) = \sum_{n=-\infty}^{+\infty} \delta(\omega - n).$$

$$Y(j\omega) = \frac{1}{2\pi}.$$

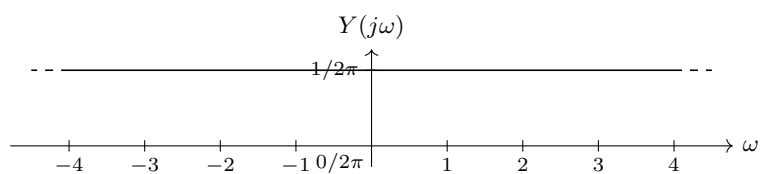
频谱图如下.



$$(8) \quad p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - 4\pi n).$$

$$P(j\omega) = \frac{1}{2} \sum_{n=-\infty}^{+\infty} \delta\left(\omega - \frac{1}{2}n\right).$$

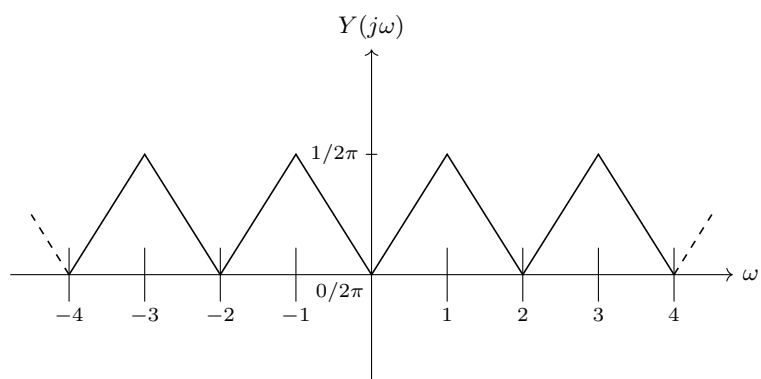
频谱图如下.



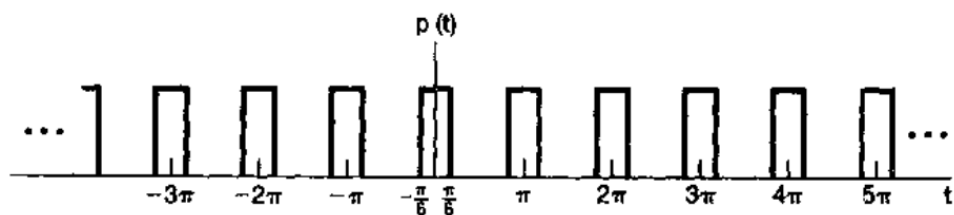
$$(9) \quad p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - 2\pi n) - \frac{1}{2} \sum_{n=-\infty}^{+\infty} \delta(t - \pi n).$$

$$P(j\omega) = \sum_{n=-\infty}^{+\infty} \delta(\omega - n) - \sum_{n=-\infty}^{+\infty} \delta(\omega - 2n).$$

频谱图如下.



(10) $p(t)$ 为下图所示周期方波.



假设方波的高度为 1.

$$p(t) = \sum_{n=-\infty}^{+\infty} \frac{1}{3} \text{Sa}\left(\frac{n\pi}{3}\right) e^{j2nt}.$$

令 $a_n = \frac{1}{3} \text{Sa}\left(\frac{n\pi}{3}\right)$, 则有

$$Y(j\omega) = \sum_{n=-\infty}^{+\infty} a_n X(j(\omega - 2n)).$$

频谱图如下.

